Chapter 1: Distributed Consensus Protocols

1.1 Distributed Transactions

Distributed transactions span multiple database nodes and require atomic commit guarantees.

1.2 Two-Phase Commit Protocol (2PC)

The Two-Phase Commit Protocol is an atomic commitment protocol for distributed databases.

Before understanding Two-Phase Commit Protocol, one must master Distributed Transactions.

Phase 1: Prepare Phase. The coordinator node sends a PREPARE message to all cohort participants.

Phase 2: Commit Phase. If all cohorts reply YES, the coordinator sends a GLOBAL-COMMIT command. If any cohort replies NO or times out, the coordinator sends a GLOBAL-ABORT command.